Model Question

Database Management System

ENCT 301

Year/Part: III/I

QN	Question	Marks	Unit
1	Explain the roles of the various layers in data abstraction along with a block diagram	3	1
2	Design an ER-model for the following mini-case, along with proper cardinality constraints. "……..Mini-case description……………" Explain referential integrity constraint in the relational model along with an illustrative example.	6+3	2
3	Consider the given relational schema ………………………. ………………….. schema description a) Write a relational algebra query for ….. (query with select/join/group by, etc) b) Write a relational algebra query for …… (database modification query)	2+2	3
4	Consider the given relational database schema ……………………. …………….. schema description a) Write an SQL query for … (query with select/joins/group by, etc) b) Write an SQL query for …. (DDL create/alter query) c) Write an SQL query for … (insert/update/delete query) d) Write an SQL query for … (view/trigger/procedure/grant query)	2*4	3
5	a) Formally define a functional dependency. Explain the Armstrong's axioms for functional dependencies. b) Explain the conditions for BCNF and 3NF. Compare them and explain which normal form would you use in which condition along with proper justification.	1+2 3+3	4
6	Describe the Pipelining execution strategy for query evaluation. Explain the key differences between Denormalization and Materialized Views, focusing on the motivation for their use and their impact on data redundancy and consistency.	2+3	5
7	a) Why is the B+ tree the preferred index structure for file organization in most modern relational database systems? b) Briefly describe the mechanism that allows Extendible/Dynamic Hashing to grow without reorganizing the entire file.	3 4	6

8	a) Define the concept of Conflict Serializability. Provide a involving two transactions (T1 and T2) that is NOT serializable and explain why it is not Conflict Serializable. b) Briefly describe the four primary states in database. Illustrate these states using a simple state diagram.	1+3 3	7
9	Describe the mechanism by which Shadow Paging ensures atomicity and durability, and state its main advantage over log-based recovery and its primary disadvantage.	2+2	8
10	Describe the two primary strategies for distributing data across multiple physical locations: Data Replication and Data Fragmentation stating a key advantage of each strategy.	2+2	9